המכון למחקרי נפט וגיאופיסיקה

THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

MAGNETO-TELLURIC MEASUREMENTS

THE TIMNA AND RAMON AREA

A. Klang

MT/665/69 July 1970

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by

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THE TIMNA AND RAMON AREA

Introduction

The first magneto-telluric measurements conducted in Israel were located in areas where no positive information on the basement is known. The drilled wells in the area remained in the sedimentary section.

An attempt was made to observe the effect of the Precambrian basement by the magneto-telluric method, known in the Eilat area. For that purpose five measurements were conducted in the Timna region, in August 1969.

In the same period a measurement was carried out near the Ramon 1 well in an attempt to obtain the depth of the basement under the cover of arkose, where the Ramon well ended.

Results

Two of the measurements, 24 and 25 , indicate the thickness of the first layer which correlates with the known sedimentary section. The following higher resistive complex should correspond to the Precambrian basement in the area. No correlation can be attempted for the deeper resistive complexes which are beyond the known geologic scale.

Data for measurement no. 26 are insufficient for lower frequencies and therefore the depth interpreted remains in the sedimentary section.

Measurements 27 and 28 indicate a relative high resistive value for the first layer. Geological work in the area indicates that the basement is at a depth of about 50 - 80 meters. (Verbal Communication, J. Bartura, Timna Copper Mines Ltd.)

The magneto-telluric method cannot differentiate between thin layers and because of this, the resistivity of the first complex represents a combined effect of the resistivity of the sedimentary section and that of the Precambrian basement section. As previously noted no correlation can be carried further for the deeper resistive complexes.

Measurement 29 located near the Ramon 1 well indicated a first low resistive complex to a depth of 500 meters followed by a very thick, more resistive complex of 15,000 meters thickness.

The first boundary is situated in the sedimentary section. The very thick second complex includes the rest of the sedimentary section followed by the arkose, penetrated by the well to a depth of 3439 meters, and a very thick unknown geologic section of about 12000 meters. This measurement showed that no basement could be differentiated under the arkose.

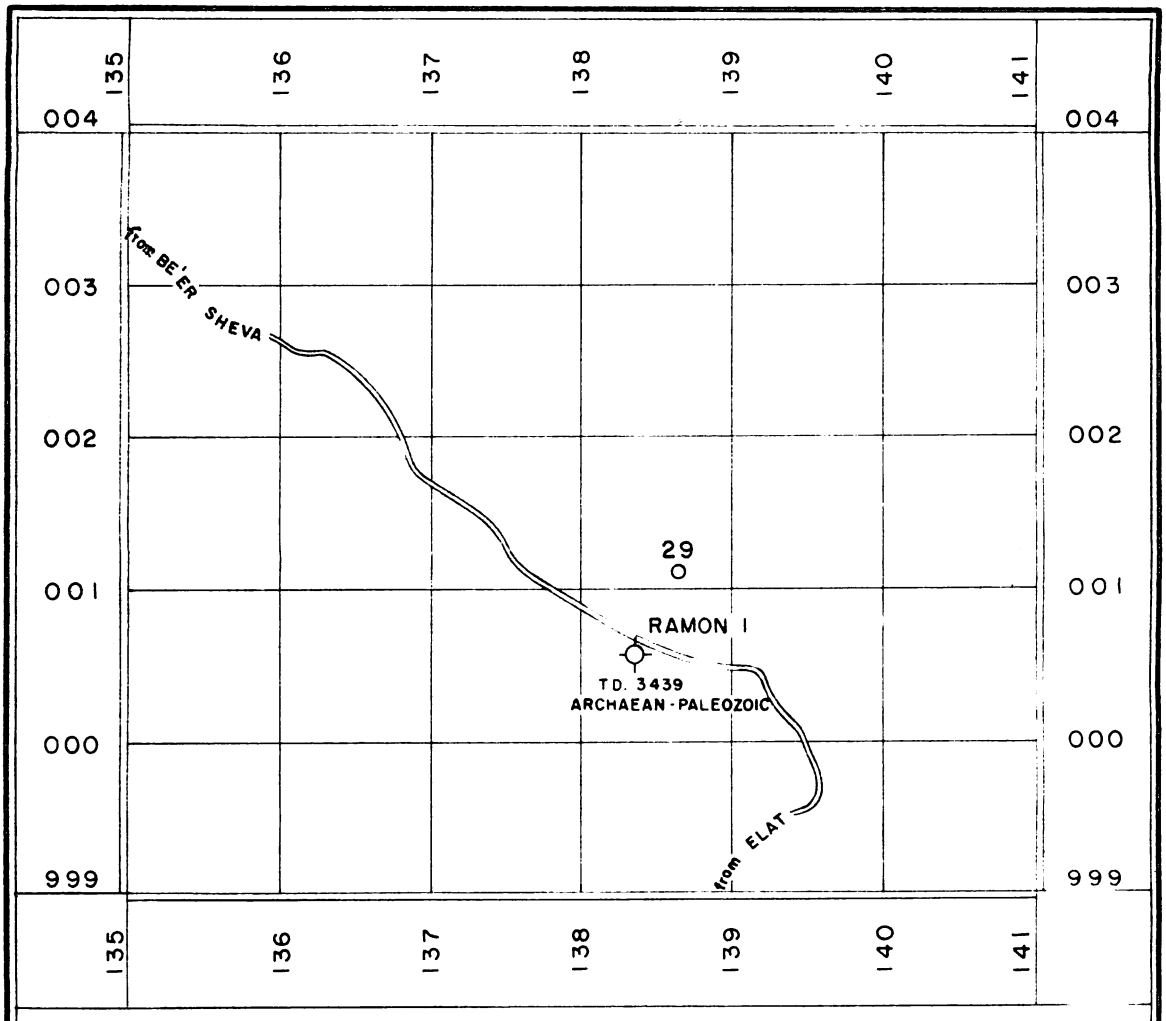
Another measurement made in the area could not be interpreted.

Conclusions

The limited number of measurements carried out in the Timna area not conclusive for the proposed problem.

It is considered that the Precambrian basement can be detected under a regular sedimentary section of at least 200-300 meter thickness, but cannot be differentiated by the resistive values under a thick cover of arkose.

Alexander Klang
Geophysicist



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MAGNETO - TELLURIC PROJECT

LOCATION MAP

SCALE

0 1 2 km.

LEGEND

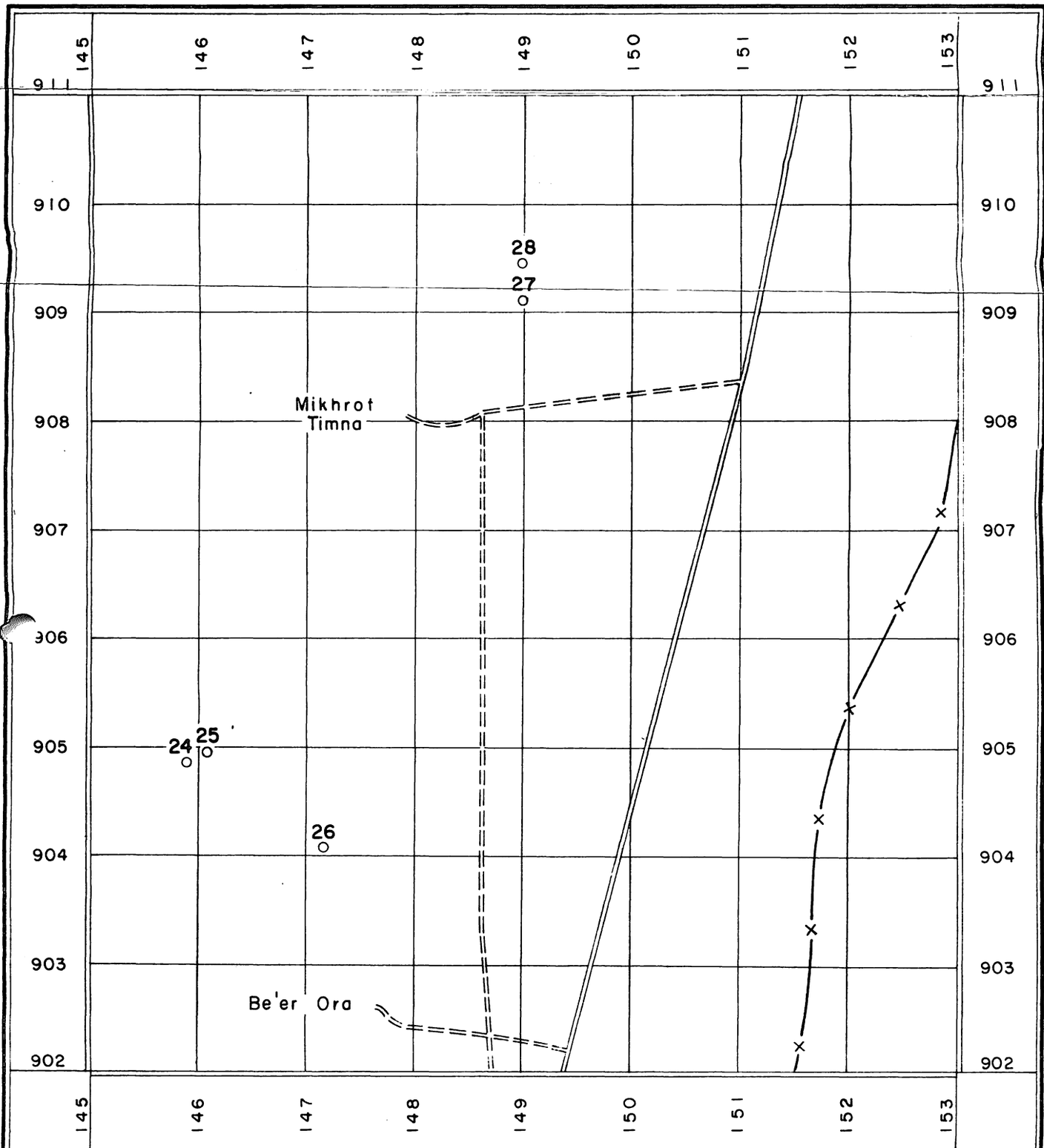
Magneto - telluric measurement

Well with total depth in metres and geologic formation penetrated

29
RAMON I
T.D. 3439
ARCHAEAN - PALEOZOIC

June 1970

MT/665/69 (8722)



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MAGNETO - TELLURIC PROJECT

LOCATION MAP

SCALE

0 1 2 km.

LEGEND

Magneto - telluric measurement 25

June 1970

MT/665/69(8721)

APPARENT RESISTIVITY Ωcm

$\sqrt{T}$ in sec.$^{\frac{1}{2}}$

CALC RES Ωcm	3000	18000	5600	10500	very high
	195	3300	9900	30000 ?	Depth in m.

M.T.P. No. 24

FILE No.

JOB No. MT/665/69 (8778)

LOCATION: TIMNA

COORDINATES: 145 85 — 904 85

ELEVATION 200 m.

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APPARENT RESISTIVITY Ωm

$\sqrt{T}$ in sec.$^{\frac{1}{2}}$

CALC. RES Ωm	3500	18000	4000	16500	very high
	300	2200	5500	38500 ?	Depth in m.

M.T.P. No. **25**

FILE No.

JOB No. **MT/665/69 (9773)**

LOCATION: **TIMNA**

COORDINATES: **146 094 — 904 973**

ELEVATION: **180 m.**

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APPARENT RESISTIVITY Ωcm.

$\sqrt{T}$ in sec.$^{\frac{1}{2}}$

CALC. RES. Ωm.	1100	8800
	200	Depth in m.

M.T.P. No. 26

FILE No. . . . . .

JOB No. MT/665/69 (8774)

LOCATION: TIMNA

COORDINATES: 147 18 — 904 10

ELEVATION 200 m.

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APPARENT RESISTIVITY Ωcm.

$\sqrt{T}$ in sec. $^{\frac{1}{2}}$

CALC. RES. Ωm.	8500	25500	5000	55000
Depth in m.	460	2760	13800	

M.T.P. No. 27..

FILE No. .....

JOB No. MT/665/69 (8775)

LOCATION: TIMNA

COORDINATES: 14895 — 90915

ELEVATION: 200 m.

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APPARENT RESISTIVITY Ωem

$\sqrt{T}$ in sec.$^{\frac{1}{2}}$

CALC RES Ωem

10000	20000	3400	67500
480	2880	10000	

Depth in m.

M.T.P. No. 28

FILE No.

JOB No. MT/665/69 (8776)

LOCATION : TIMNA

COORDINATES : 14895 — 90945

ELEVATION 205 m.

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APPARENT RESISTIVITY ohm.m

$\sqrt{T}$ in sec.$^{\frac{1}{2}}$

CALC. RES. ohm.m

1100 | 7700 | 70000

500 | 15500 | Depth in m.

M.T.P. No. 29

FILE No. . . . . .

JOB No. MT/665/69 (8777)

LOCATION: RAMON

COORDINATES: 13865 — 00115

ELEVATION 500 m.

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